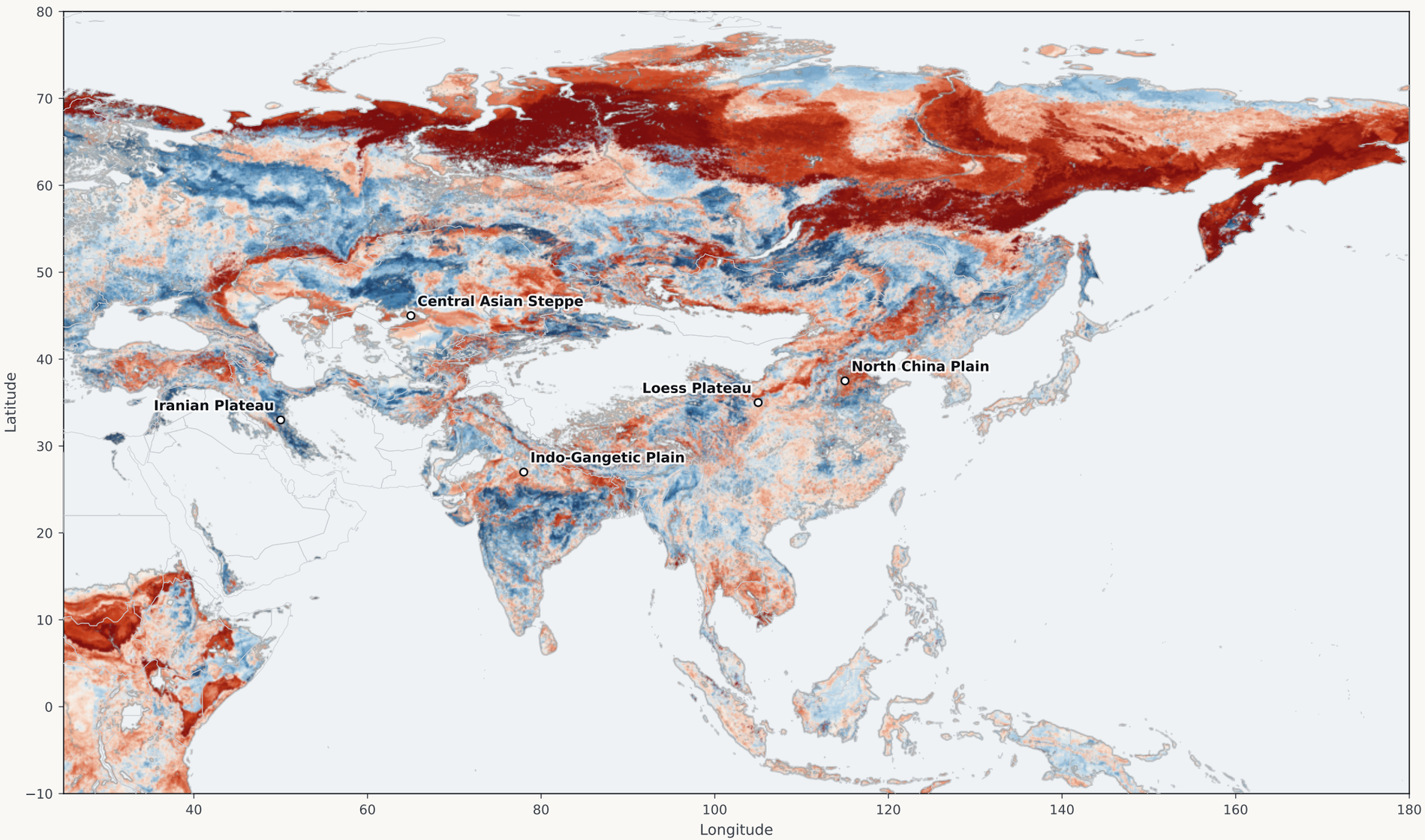
MSHI-GEO / ASIA / CLIMATE + NPP MODEL

Climate-Corrected Soil Microbial Respiration Anomaly

Predicted soil respiration ÷ climate-expected soil respiration. Values below 1.0 indicate microbial activity is suppressed relative to what climate alone would support — a signal of soil-driven degradation.
Adds MODIS satellite vegetation productivity — best transfer of any config



ANOMALY RATIO

INTERPRETATION

≥ 1.25	Microbial activity exceeds climate expectation
1.00 – 1.25	Healthy / functioning soils
≈ 1.00	At climate expectation
0.75 – 1.00	Mild suppression
≤ 0.75	Significant degradation — addressable by management

MODEL XGBoost · spatial-block CV · resolution ~5 km
DATA COSORE + SRDB soil respiration · SoilGrids 2.0 · WorldClim 2.1 · MODIS

CV R^2 -0.021 Asia→US R^2 0.145
N(train)=463 N(US)=223 2026-05-07